National University of Computer and Emerging Sciences, Lahore Campus

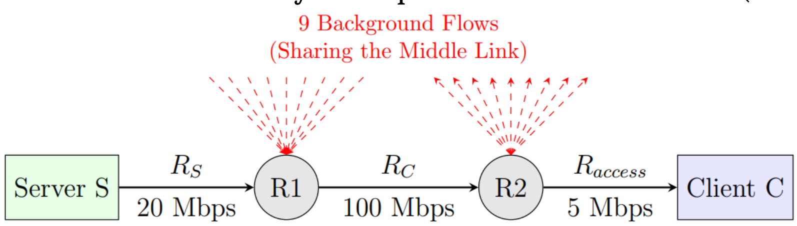
	Course Name:	**Computer Networks**	**Course Code:**	**CS 3001**
	Program:	**BS (Computer Science)**	**Semester:**	**Spring 2026**
	Duration:	**15 minutes**	**Total Marks:**	**15**
	Paper Date:		**Section**	**6B1**
	Exam Type:	**Quiz 1 - Chapter 1**	**Page(s):**	**2**

Student Name **Roll No.** **Section:**

Q1. Carefully read the Question before attempting: **[5 marks]**

Consider the file transfer scenario shown below. Server S sends a file to Client C through two intermediate routers R1 and R2.

- Link S → R1: Rate R_S = 20 Mbps.
- Link R1 → R2: Shared link, Rate R_C = 100 Mbps. This link is shared by 9 other background traffic flows (assume all are working at full capacity).
- Link R2 → C: Rate R_{access} = 5 Mbps.
- The 9 background flows consume exactly 10 Mbps each of the middle link (R1 → R2).

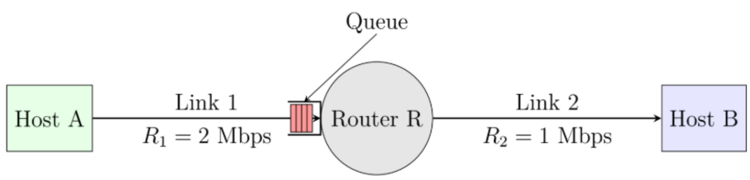


Calculate the throughput for the file transfer from S to C. Show the available bandwidth for each link and identify the bottleneck link.

Q2. Carefully read the Question before attempting: **[10 marks]**

Consider a packet-switched network where Host A sends a packet to Host B via a single Router R.

- Link 1 (A → R): Transmission Rate R1 = 2 Mbps. Propagation Delay d_{prop1} = 10 ms.
- Link 2 (R → B): Transmission Rate R2 = 1 Mbps. Propagation Delay d_{prop2} = 10 ms.
- Packet Size: All packets have a length L = 2000 bits.
- Processing Delay: Negligible at Router R ($d_{proc} \approx 0$).
- Queuing: Router R has a buffer. Since R1 > R2, packets arriving at R faster than they can be transmitted to B, may cause a queue to build up.



1. Calculate the transmission delays at each Link?
2. What is the total end to end delay considering negligible queuing and processing delays (router must receive the whole packet before starting transmission)?